Assignment 1.

1 . A.
Colab link -
https://colab.research.google.com/drive/1AaCwyugH5epCKNqxlgBDH-UVqz6GXmUp#scrollTo=TeGfskiVmMhK(the file contain code for below experiment)

Dataset=MNIST , Epoch = 8

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	99	98.06
16	**SGD**	**0.0001**	95.06	91.4
16	**Adam**	**0.001**	98.33	97.93
16	**Adam**	**0.0001**	98.83	97.43
32	**SGD**	**0.001**	97.43	97.93
32 [Optional]	**SGD**	**0.0001**		
32 [Optional]	**Adam**	**0.001**		
32	**Adam**	**0.0001**	95.03	93.12

Dataset=FashionMNIST , Epoch = 8

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	87.76	86.76
16	**SGD**	**0.0001**	79	74.13
16	**Adam**	**0.001**	89.5	88.5

16	**Adam**	**0.0001**	89.9	88.56
32	**SGD**	**0.001**	85.8	82.86
32 [Optional]	**SGD**	**0.0001**		
32 [Optional]	**Adam**	**0.001**		
32	**Adam**	**0.0001**	88.43	86.2

Dataset=MNIST , Epoch = 1 , with vary hyper parameter.

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	97.53	97.49
16	**SGD**	**0.0001**	87.60	53.1
16	**Adam**	**0.001**	97.25	95.86
16	**Adam**	**0.0001**	98.9	98.64
32	**SGD**	**0.001**	96.36	96.81
32 [Optional]	**SGD**	**0.0001**		
32 [Optional]	**Adam**	**0.001**		
32	**Adam**	**0.0001**	97.85	97.925

Dataset=FashionMNIST , Epoch = 1 , with vary hyper parameter.

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	82.75	82.01

16	**SGD**	**0.0001**	75.91	66.69
16	**Adam**	**0.001**	86.13	82.96
16	**Adam**	**0.0001**	88.01	79.06
32	**SGD**	**0.001**	81.37	79.04
32 [Optional]	**SGD**	**0.0001**		
32 [Optional]	**Adam**	**0.001**		
32	**Adam**	**0.0001**	88.375	87.725

Detailed Analysis -

1. ResNet-18 performs better than ResNet-50, despite having fewer parameters. This suggests that increasing model depth does not improve performance and may even introduce overfitting.
2. Adam optimizer shows more stable and higher accuracy.Adam adapts learning rates automatically for each parameter, which helps faster and more reliable convergence in low-epoch training scenarios like this.
3. A learning rate of 0.001 provides a good balance between convergence speed and stability for MNIST, especially under limited epochs.
4. Batch size 16 generally performs slightly better than batch size 32.smaller batch sizes introduce beneficial gradient noise, which can improve generalization on simpler datasets.

1. B.

	Dataset	Kernel	C	Degree	Test Accuracy (%)	Training Time (ms)
0	MNIST	rbf	1.0	NaN	94.45	13825.66
1	MNIST	rbf	10.0	NaN	95.55	7533.72

2	MNIST	poly	1.0	2.0	94.15	6714.92
3	MNIST	poly	1.0	3.0	93.40	7622.94
4	FashionMNIST	rbf	1.0	NaN	86.00	7736.48
5	FashionMNIST	rbf	10.0	NaN	87.35	6830.12
6	FashionMNIST	poly	1.0	2.0	84.75	6979.44
7	FashionMNIST	poly	1.0	3.0	82.80	8153.68

2.

				Test Classification Accuracy (in %)			**Train Time (in ms)**			**FLOPs**		
Compute	**Batch Size**	**Optimizer**	**Learning Rate**	**ResNet-18**	**ResNet-32 [Optional]**	**ResNet-50**	**ResNet-18**	**ResNet-32 [Optional]**	**ResNet-50**	**ResNet-18**	**ResNet-32 [Optional]**	**ResNet-50**
CPU	**16**	**SGD**	**0.001**	80.35		62.20	161201.66		352667.72	33097472		78353152
CPU	**16**	**Adam**	**0.001**	79.85		61.70	219992.64		456093.67	33097472		78353152
GPU	**16**	**SGD**	**0.001**	78.55		61.00	7872.07		18305.87	33097472		78353152
GPU	**16**	**Adam**	**0.001**	77.00		62.40	9293.69		17419.21	33097472		78353152

Detail analysis.

1 ResNet18 consistently achieves the highest accuracy among the tested models, reaching approximately 80% accuracy on CPU and 78–79% on GPU.ResNet50, despite being significantly deeper, shows lower accuracy (~61–62%).This indicates that

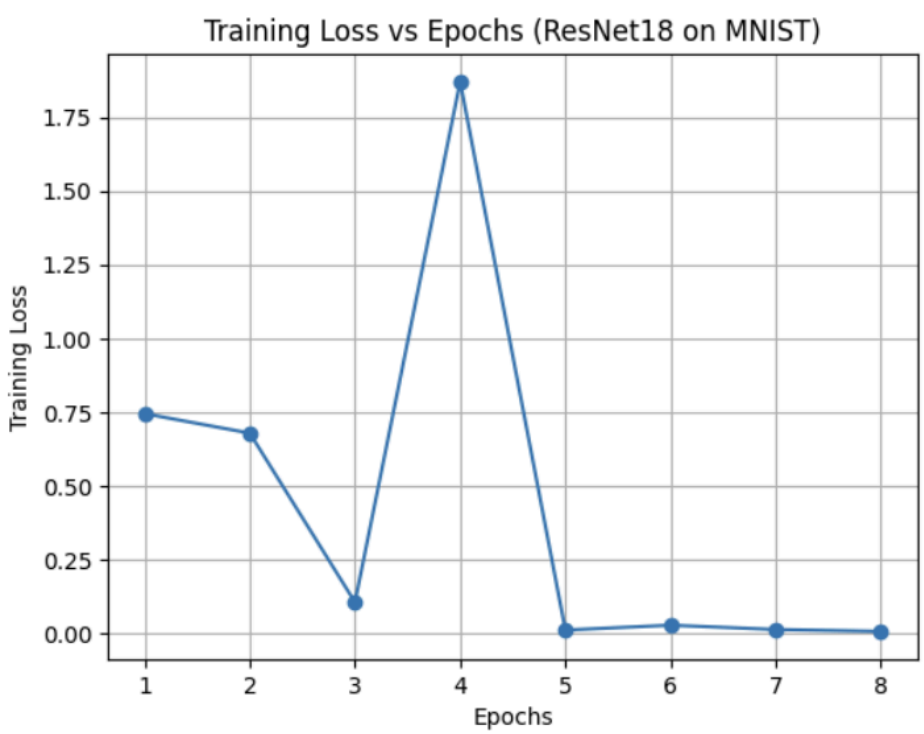
increasing model depth does not necessarily improve performance for FashionMNIST, which is a relatively simple dataset.For small datasets with limited complexity, SGD with a fixed learning rate often generalizes better than adaptive optimizers like Adam.

2 GPU acceleration is more effective for deeper networks.For smaller models like ResNet18, GPU overhead reduces the relative speed-up

3 FLOPs measure theoretical computational complexity, not execution speed.FLOPs remain constant across CPU/GPU and optimizers, since they depend only on Network architecture and Input resolution

Results -

1. Best Model for MNIST dataset -
   Model name: ResNet18,
   optimizer: SGD,
   Learning rate: 0.001,
   Batch size: 16,
   Test accuracy : 99



2. Best Model for FashionMNIST dataset -
   Model name: ResNet18,
   optimizer: ADAM,
   Learning rate: 0.0001,

Batch size: 16,
Test accuracy : 89.9